

Kaushik Shroff

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EDUCATION

University of Wisconsin-Madison

Aug 2024 – Dec 2025

Master of Science, Electrical and Computer Engineering

GPA: 3.5/4.0

- **Coursework:** Advanced Computer Architecture, Operating Systems, VLSI Systems Design

Savitribai Phule Pune University

Jul 2020 – Jun 2024

Bachelor of Engineering in Electronics and Telecommunication

GPA: 3.8/4.0

- **Coursework:** Digital Circuits, Microcontrollers, VLSI

WORK EXPERIENCE

Research Assistant — Chiplet Interconnect Design for Mamba LLM Accelerator

May 2025 – Aug 2025

University of Wisconsin-Madison

Madison, WI

Tools: Verilog, SystemVerilog, BookSim, SET, Python

- Collaborating with a **PhD Researcher** on chiplet-based hardware acceleration for the Mamba large language model (LLM).
- Integrated **Floret** SFC into SET (8x8, 64 chiplets) and replaced mesh with hierarchical unicast/multicast cost models.
- Achieved **24.6% lower latency** and **55.6% lower DRAM energy** vs. mesh, with a **+139.4% NoC energy** tradeoff.
- Implemented **BF16 Huffman compression** with 32-bit flit packing and dual-phase evaluation (raw vs. compressed).
- Reduced BODY flits by **65%**, cut link transfer latency by **61%**, achieving an overall **2.6x speedup**.

PROJECTS

Out Of Order CPU Power Modeling in gem5

Tools: gem5, McPAT, CACTI, Python, C++

- Built a **power model** for gem5's **Out Of Order CPU**, enabling power-performance analysis of out-of-order pipelines.
- Mapped **20+ micro-architectural components** (e.g., ROB, branch predictor, LSQ) to activation statistics in gem5.
- Extracted **per component energy** using McPAT, improving the modeling accuracy for ALU, caches, and branch units.
- **Automated energy computation** via Python scripts to parse statistics, producing component-wise power with **98% accuracy**.
- Validated model by **benchmarking 7 workloads**, observing 2x higher power for integer-intensive code.
- Extended gem5 by **adding 5+ custom stat counters** and integrated support for segment-based power analysis.

MUSI Cache Coherence Protocol Implementation in gem5

Tools: gem5, SLICC, C++, Python

- Implemented the COUP concept by extending gem5's MSI protocol into a **MUSI (Modified-Update-Shared-Invalid)** variant.
- Added **5+ new coherence messages** and **10+ state transitions** across cache and directory controllers using SLICC.
- Designed and integrated a **new state of update (U)** to support commutative updates, enabling local aggregation.
- Built **custom C++ micro-benchmarks** to test COUP behavior under conflicting shared-memory access patterns.
- Identified and resolved protocol-level integration issues by instrumenting **fine-grained debug tracing** in cache hierarchy.
- Verified **syntactic and build correctness** of protocol extension.

Graph Neural Network (GNN) Accelerator Design

Tools: SystemVerilog, ModelSim, Design Compiler, Innovus, PrimeTime, Virtuoso, Tcl

- Designed a **GNN accelerator** with MAC, ReLU, and aggregation units for 4-task scheduling on heterogeneous Arm cores.
- Achieved 869 MHz max frequency, 3.07 mW power, and **7.91 pJ·ns·mm² EDAP** through optimization techniques including **clock gating, bit-width pruning, and modular pipelining**.
- Synthesized, placed, and routed using 7nm standard-cell libraries, followed by post-layout **power and timing analysis**.
- **Reduced post-APR power by 5% and pre-APR by 38% using clock gating**, demonstrating hardware-aware optimization.
- Completed **full frontend-to-backend development**, from RTL design to physical layout and post-silicon estimation, including verification, synthesis, APR and power/timing closure.

Scan-Based Test Generation for Digital Systems

Tools: Synopsys Design Vision, TetraMax, Tcl, VHDL

- Generated ATPG test sets for full and partial scan designs of a Viper processor (ITC'99 b14 benchmark), achieving **99.48% test coverage** with full scan.
- Designed and synthesized partial scan architectures using **SCOAP-based flip-flop ranking**, optimizing for test coverage, area, and test time.
- Conducted design space exploration across 17 configurations, achieving a **122% improvement** in test efficiency over full scan.
- **Automated** scan chain generation, test compression, and metric evaluation using custom **Tcl scripting** and analysis workflows.

SKILLS

- **Languages:** Python, C++, C, SLICC, Bash, Tcl
- **HDL:** Verilog, SystemVerilog, UVM, VHDL
- **Tools:** gem5, McPAT, CACTI, BookSim, SET, Design Compiler, ModelSim, Innovus, PrimeTime, Virtuoso, Git